PHYS205 GENERAL PHYSICS III
FALL 2025

Instructor: Aşkın Kocabaş
Class Meeting Time and Location: Tue, Thu 10-11:10 Sci Z24
PS Meeting Time and Location:

Office Hours: Tue 9:00
Email: akocabas@ku.edu.tr
TA: Zeynep Sena Ceyhan, Ömer Fatih Dokumacı
Number of KU credits: 3
Language: English

Course description
PHYS205 continues PHYS101 with a more detailed look at oscillations and waves, and also covers the basics of thermodynamics and thermal physics. We will spend the first half of the course to understand oscillating systems of various kinds, including coupled oscillations and their continuous limit, waves. Without exaggeration, oscillations are the heart of our modern understanding of physics from classical mechanics. A solid grasp of these topics is essential for understanding nature. We will cover the atomic basis of heat and thermal phenomena in the second half. These topics are fascinating, and they led to our current theories of the structure of matter. You will have 6 laboratory sessions as part of the class. Please contact Nazmi Yilmaz for inquiries about the labs.

Textbook: You will have lecture notes for all topics. I will be following parts of *Waves: Berkeley Physics Course Vol. 3* for oscillations, and *University Physics* (PHYS101-102 textbook) for thermodynamics.

Learning Objectives
Upon successful completion of this course, we will learn
1. the rich phenomena of oscillations and their ubiquity in explaining everything from quarks to the whole universe
2. the behavior of fluids, sound waves and electromagnetic waves
3. the underlying mechanism of thermal phenomena such as flow of heat and behavior of gases
4. the fundamental role of the first and second law of thermodynamics to understand matter
5. how to repair your broken refrigerator
6. thermal radiation

Assessment methods & grading scheme:

Type	**Final Grade %**
Problem Sets	10
Midterm 1 & 2	40
Final	30
Lab	15
Attendance	5
Total	**100**

Grading: We will follow the regular curve-based grading policy.

Problem Sets: You will have weekly problem set assignments. *At least 25% of the questions in the exams will be identical or almost identical versions of the homework questions.*

Makeup Policy: If you miss an exam and have an excuse, it is your responsibility to officiate this through the Health Center. **There will be a single makeup exam after the finals for any exam(s) you miss. This makeup exam will cover all topics. There will be no makeup for problem sets.**

Tentative Syllabus:
Week 1: Simple Harmonic Motion
Week 2: Damped and Forced Oscillations
Week 3: Coupled Oscillations
Week 4: Coupled Oscillations
Week 5: Waves: Oscillations of continuous systems
Week 6: Waves: Oscillations of continuous systems
Week 7: Waves: Oscillations of continuous systems
Week 8: Fluid Mechanics and Sound Waves
Week 9: Temperature and Heat
Week 10: Thermal Properties of Matter
Week 11: Thermal Properties of Matter
Week 12: First Law of Thermodynamics
Week 13: Second Law of Thermodynamics
Week 14: thermal radiation

We might make small changes to the above plan depending on student interest and our pace.

Koç University
Statement on Academic Honesty
with Emphasis on Plagiarism

Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct (https://apdd.ku.edu.tr/en/academic-policies/student-code-of-conduct/). Actions considered as academic dishonesty at Koç University include but are not limited to cheating, plagiarism, collusion, and impersonating. This statement's goal is to draw attention to cheating and plagiarism related actions deemed unacceptable within the context of Student Code of Conduct:

All individual assignments must be completed by the student himself/herself, and all team assignments must be completed by the members of the team, without the aid of other individuals. If a team member does not contribute to the written documents or participate in the activities of the team, his/her name should not appear on the work submitted for evaluation.

Plagiarism is defined as 'borrowing or using someone else's written statements or ideas without giving written acknowledgement to the author'. Students are encouraged to conduct research beyond the course material, but they must not use any documents prepared by current or previous students, or notes prepared by instructors at Koç University or other universities without properly citing the source. Furthermore, students are expected to adhere to the Classroom Code of Conduct (https://apdd.ku.edu.tr/en/classroom-code-of-conduct/) and to refrain from all forms of unacceptable behavior during lectures. Failure to adhere to expected behavior may result in disciplinary action.

There are two kinds of plagiarism: Intentional and accidental. Intentional plagiarism (Example: Using a classmate's homework as one's own because the student does not want to spend time working on that homework) is considered intellectual theft, and there is no need to emphasize the wrongfulness of this act. Accidental plagiarism, on the other hand, may be considered as a 'more acceptable' form of plagiarism by some students, which is certainly not how it is perceived by the University administration and faculty. The student is responsible from properly citing a source if he/she is making use of another person's work. For an example on accidental plagiarism, please refer to the document titled "An Example on Accidental Plagiarism".

If you are unsure whether the action you will take would be a violation of Koç University's Student Code of Conduct, please consult with your instructor before taking that action.